

I develop machine learning methods that turn complex learned representations into human-understandable concepts and structure for scientific discovery. My recent work focuses on discovering reusable concepts from foundation-model representations, using them to generate new scientific insights and make model-discovered structure accessible to domain experts. I have applied these ideas to biomedical imaging, critical-care EEG, and wearable cardiac monitoring, with work published in NEJM AI, Nature Machine Intelligence, JMLR, CVPR, and Harvard Data Science Review.

EDUCATION	Duke University Postdoctoral Researcher Ph.D. in Electrical and Computer Engineering (Advisor: Cynthia Rudin) M.S. in Computer Science (Transferred to Ph.D. Program)	Durham, NC July 2022 – Now Jan 2022 – May 2026 Aug 2020 – Dec 2021
	Rensselaer Polytechnic Institute B.Sc. in Computer Science	Troy, NY July 2016 – May 2020

PUBLICATIONS (* indicates co-first / co-senior authors, equal contribution) [Google Scholar](#)

- [1] “Data-hugging shields proprietary AI models from research that could disprove them.” Anish Karpurapu, **Zhicheng Guo**, Xiao Hu, Cynthia Rudin.
npj Artificial Intelligence, 2026. 
- [2] “ProtoCDisco: Unsupervised Concept Discovery using Foundation Model Representations”
Zhicheng Guo, Maximillian Machado, Jon Donnelly, Cynthia Rudin. In Submission
- [3] “Cosine Similarity is Almost all you Need (for Prototypical-Part Models).” Luke Moffett, Frank Willard, Maximillian Machado, Emmanuel Mokel, Jon Donnelly, **Zhicheng Guo**, Adam Costarino, Julia Yang, Giyoung Kim, Alina Jade Barnett, Cynthia Rudin.
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026. 
- [4] “What is Different Between These Datasets? A Framework for Explaining Data Distribution Shifts.” Varun Babbar*, **Zhicheng Guo***, Cynthia Rudin.
Journal of Machine Learning Research (JMLR), 2025. 
- [5] “SiamAF: Improving atrial fibrillation detection using a shared latent space for ECG and PPG signals.” **Zhicheng Guo**, Cheng Ding, Duc H. Do, Amit Shah, Randall J Lee, Xiao Hu, Cynthia Rudin.
Harvard Data Science Review (HDSR), 2025. 
- [6] “It’s LIT! LLMs with Interpretable Tools.” Ruixin Zhang, Jon Donnelly, **Zhicheng Guo**, Ghazal Khalighinejad, Haiyang Huang, Alina Jade Barnett, Cynthia Rudin.
Multi-Turn Interactions in Large Language Models Workshop (MTI-LLM Workshop, NeurIPS), 2025. 
- [7] “Rashomon Sets for Prototypical-Part Networks: Editing Interpretable Models in Real-Time.” Jon Donnelly, **Zhicheng Guo**, Alina Jade Barnett, Hayden McTavish, Chaofan Chen, Cynthia Rudin.
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025. 
- [8] “Improving Clinician Performance in Classification of EEG Patterns on the Ictal-Interictal-Injury Continuum using Interpretable Machine Learning.” Alina Jade Barnett*, **Zhicheng Guo***, Jin Jing*, Wendong Ge, Brandon Westover, Cynthia Rudin.

New England Journal of Medicine AI (NEJM AI), 2024. 

- [9] “Learning From Alarms: A Robust Learning Approach for Accurate Photoplethysmography-Based Atrial Fibrillation Detection Using Eight Million Samples Labeled with Imprecise Arrhythmia Alarms.” Cheng Ding, **Zhicheng Guo**, Cynthia Rudin, Ran Xiao, Amit Shah, Duc H Do, Randall J Lee, Gari Clifford, Fadi B Nahab, Xiao Hu. *IEEE Journal of Biomedical and Health Informatics (JBHI)*, 2024. 
- [10] “Learned Kernels for Sparse, Interpretable, and Efficient Medical Time Series Processing.” Sully F Chen, **Zhicheng Guo**, Cheng Ding, Xiao Hu, Cynthia Rudin. *Nature Machine Intelligence*, 2024. 
- [11] “SiamQuality: a ConvNet-based Foundation Model for Photoplethysmography Signals.” Cheng Ding, **Zhicheng Guo**, Zhaoliang Chen, Randall J Lee, Cynthia Rudin, Xiao Hu. *Physiological Measurement*, 2024. 
- [12] “Integrated Single-cell Multiomic Analysis of HIV Latency Reversal Reveals Novel Regulators of Viral Reactivation.” Manickam Ashokkumar, Wenwen Mei, Jackson J Peterson, Yuriko Harigaya, David M Murdoch, David M Margolis, Caleb Kornfein, Alex Oesterling, **Zhicheng Guo**, Cynthia D Rudin, Yuchao Jiang, Edward P Browne. *Genomics, Proteomics & Bioinformatics*, 2024.
- [13] “A Supervised Machine Learning Semantic Segmentation Approach for Detecting Artifacts in Plethysmography Signals from wearables.” **Zhicheng Guo**, Cheng Ding, Xiao Hu, Cynthia Rudin. *Physiological Measurement*, 2021. 

PROFESSIONAL
EXPERIENCE

Duke University

Jul 2026 – Present

Postdoctoral Researcher — Foundation Models & Interpretability

- Developing vision foundation-model and representation-learning methods for breast cancer risk prediction from large-scale mammography data.
- Building an interpretable lexicon of mammographic texture concepts, together with scalable evaluation pipelines for high-stakes medical imaging.
- Developing unsupervised concept-discovery methods from vision foundation-model representations and evaluating the discovered concepts through human studies.

Texas Instruments – Kilby Labs

May 2025 – Aug 2025

Machine Learning Researcher Intern

- Architected and deployed a self-evolving LLM-agentic optimization framework for hardware-constrained neural architecture search and mixed-precision quantization on edge hardware.
- Implemented multi-objective reward functions and on-the-fly hardware profiling to guide optimization decisions and improve real-time performance.
- Developed specialized small reasoning language model groups (SLMs) as replacements for generic LLM agents with GRPO and finetuning, enabling on-site and data-secure inference at a fraction of compute cost.

Duke University Health System

May 2021 – Aug 2021

Quantitative Research Intern

- Investigated correlations between cellular populations/markers in CSF and PBMC with cognitive scores in HIV-associated neurocognitive disorder patients under different drug use conditions.
- Developed machine learning models with flow cytometry data to predict drug use and impairment, enabling clinical researchers to assess drug effects on patients.

PATENTS **Texas Instruments – Kilby Labs**

- [1] LLM-guided neural network architecture and quantization search for TINIE (Texas Instruments Neural Inference Engine) on microcontrollers. (pending)
- [2] Small language models for TINIE-optimized edge AI model development. (pending)

MEDIA **OP-EDS**

- [1] Cynthia Rudin, **Zhicheng Guo**, Cheng Ding, and Xiao Hu. “How good are AI health technologies? We have no idea.” *STAT News*, Oct. 11, 2023. 

News Articles

- [2] “AI helps medical professionals read confusing brain EEGs” *Mindplex*, May 29, 2024. 
- [3] “New AI tool from Duke researchers reads EEGs, could improve treatment of seizures” *Yahoo News (The News&Observer)*, Feb. 20, 2025. 

SERVICES

Reviewer

IEEE Journal of Biomedical and Health Informatics	2023, 2024, 2025
Physiological Measurement	2025
Epilepsia	2024
NeurIPS	2024, 2025, 2026

TEACHING

Duke University

Graduate Theory and Algorithms for Machine Learning 671D	Fall 2025
- Lecture on neural networks and Interpretable Neural Networks Design	
TA, Graduate Theory and Algorithms for Machine Learning 671D	Fall 2023
TA, Graduate Theory and Algorithms for Machine Learning 671D	Fall 2022
TA, Graduate Theory and Algorithms for Machine Learning 671D	Fall 2021

Rensselaer Polytechnic Institute

SAPRIS Program Academic Mentor	Summer 2017
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TALKS

INFORMS

Invited talk on Interpretable Deep Learning Models	Atlanta, GA October 2025
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Cornell ORIE Young Researchers Workshop

Invited poster	Ithaca, NY October 2024
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ADVISED STUDENTS

Will Li — Master’s student, Duke University

Anish Karpurapu — MD-PhD student, Duke University

Maximillian Machado — Master’s student, incoming PhD student, Duke University

Liam Connor — Master’s student, Duke University

Pranay Jain — Former master’s student, now PhD student, University of Washington

Haoyu (Cindy) Wu — Former master’s student, now in industry

Dennis Tang — Undergraduate student, incoming PhD student, Duke University

Alex Oesterling — Former undergraduate student, now PhD student, Harvard University

Edna Zhang — Former master’s student, now in industry

Adam Costarino — Former master’s student, now in industry

Bhrij Patel — Former undergraduate student, now PhD student, University of Maryland